

Fizika

9-sinf

Laboratoriya ishi №1

Mavzu: **Qattiq jismlarning solishtirma issiqlik sig‘imini aniqlash**

Dars rejasi

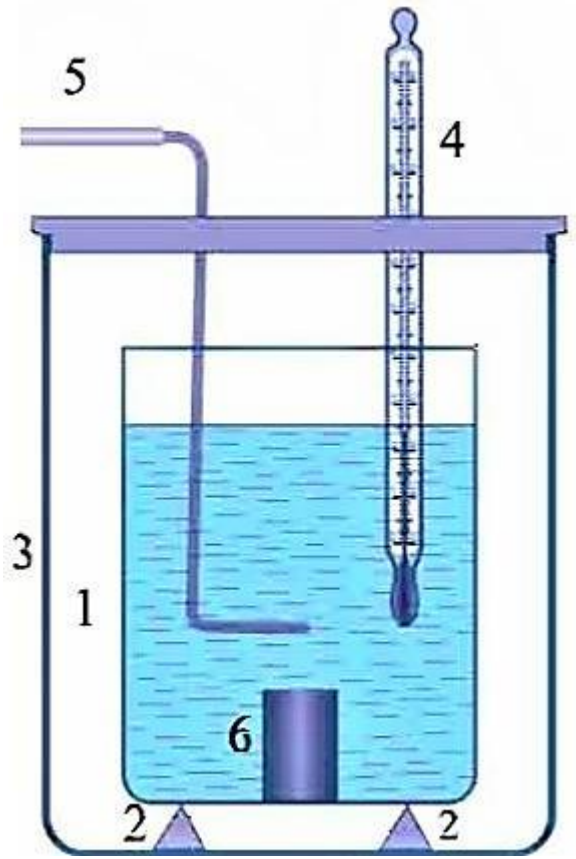
Laboratoriya ishini bajarishda texnika xavfsizligi qoidalari bilan tanishish

Laboratoriya ishini bajarishda kerakli bo‘lgan jihozlar bilan tanishish

Laboratoriya ishini tajribada ko‘rsatish

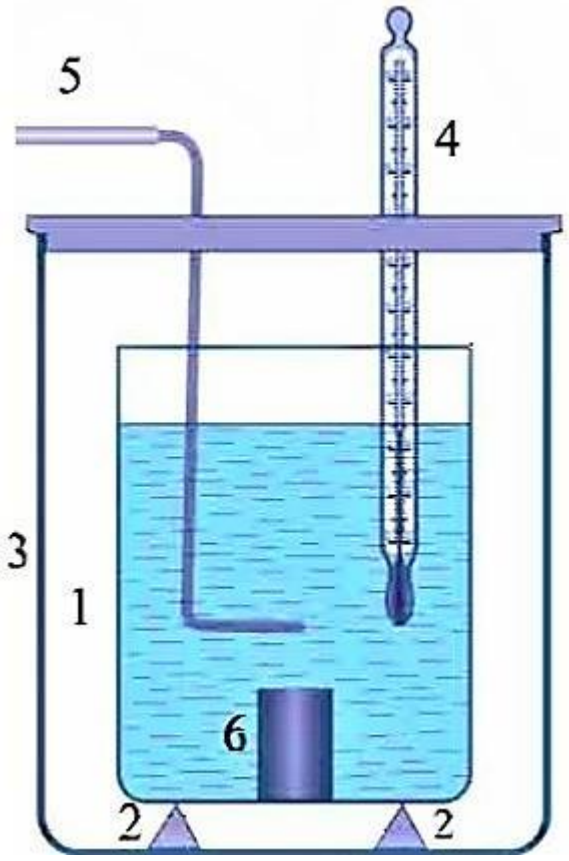
Ishning maqsadi

Jismlarning solishtirma issiqlik sig‘imini aniqlash.



Kerakli asbob va jihozlar

- Kalorimetr va aralashtirgichi;
- Tarozi;
- Termometr;
- Solishtirma issiqlik sig‘imi aniqlanadigan 3 ta bir xil moddadan tayyorlangan turli xil massadagi jismlar;
- Qaynoq suv.



Ishni bajarish tartibi

1. Kalorimetr va aralashtirgichni birgalikda torozida tortib, ularning massasini aniqlang (m_k). $c_k = 890 \text{J/kg}^0\text{C}$.

2. Menzurka yordamida suv hajmini (V) o‘lchab, uni kalorimetr idishiga quying.

3. Kalorimetrga quyilgan suvning massasini $m_s = \rho_s V_s$ formuladan foydalanib hisoblang.

4. Kalorimetrga termometrni tushiring. Biroz kuting. Issiqlik muvozanati yuzaga kelganda suvning temperaturasini (t_s) aniqlang.

5. Solishtirma issiqlik sig‘imi aniqlanayotgan jismning massasini (m_j) tarozida o‘lchang.

Ishni bajarish tartibi

6. Jismni ipga bog‘lab qaynoq suv ichiga tushiring. Jism va suv o‘rtasida issiqlik muvozanati vujudga kelishini kuting. Qaynoq suvning temperaturasini (t_s) termometr yordamida o‘lchab oling.

7. Qaynoq suvdan olingan jismni tezlik bilan sovuq suv solingan kalorimetr ichiga tushiring. Aralashtirgich bilan kalorimetrdagi suvni aralashtiring va termometr ko‘rsatgan aralashmaning (t_a) temperaturasini yozib oling.

Ishni bajarish tartibi

8. Quyidagi formula yordamida jismning solishtirma issiqlik sig‘imini aniqlang:

$$c = \frac{(m_s \cdot c_s + m_k \cdot c_k) \cdot (t_a - t_s)}{m_j (t_j - t_a)}$$

9. Tajribani 3 marta takrorlang va solishtirma issiqlik sig‘imlari uchun o‘rtacha qiymatini hisoblang.

10. Olingan natijalarni jadvalga yozing.

O‘lchash va hisoblash natijalari

№	m_k, kg	m_s, kg	m_j, kg	c_k,J/kg·°C	t_s,°C	t_j, °C	t_a, °C	c_j,J/kg·°C	$c_{j,o‘rt}$
1									
2									
3									

Mustaqil bajarish uchun topshiriqlar

1. Solishtirma issiqlik sig‘imining fizik ma’nosini yozing.
2. Issiqlik muvozanati deb nimaga aytiladi?
3. O‘lchash natijalarini tahlil qiling va xulosa yozing.